

Tunable Transparency Metamaterials Based on Liquid Crystals

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It is shown that one can design a metamaterial slab that has high optical transparency. The slab is composed of randomly distributed solid nanospheres in a liquid crystal film, whose properties are determined using the classical Drude and Mie theories. A genetic algorithm optimization is used to determine the nanosphere size and density in the slab such that it is transparent to optical frequencies, but can be tuned to become opaque by changing the permittivity of the liquid crystal.

1. INTRODUCTION

A major aspiration of material science is to construct materials with prescribed properties, such as the response to electromagnetic radiation, static electric and magnetic fields and various other forces. This endeavor involves the description of the structure and properties of matter at distinct scales, the microscopic (atomic and molecular), the mesoscopic (nanoscale) and the macroscopic. For a given material the macroscopic (bulk) parameters are determined by averaging over the mesoscopic parameters, which are in turn, determined by averaging over the microscopic ones. Ultimately the properties of a material are determined by its microscopic character; however a wide range of bulk behavior can be produced by nanoscale mixtures in the form of clusters, layers and/or deposited patterns. Of course each component derives its attributes from its microscopic constituents but the mesoscopic design can significantly control the values of specific bulk properties. In order to describe the behavior of the microscopic structure one must utilize quantum mechanical procedures, such as Time Dependent Density Functional and Quantum mechanical Greens Function [?] methods. However, these procedures are very expensive in terms of computer power and time and thus not practical tools with which to explore changes in macroscopic characteristics.

The details of the interaction between electromagnetic radiation and materials are embodied in the frequency dependent dielectric response function, which in general, has a complicated functional form containing multiple poles and branch cuts. However nanosize clusters, wires or slabs can be effectively and accurately described by rather simple response functions in the classical Drude picture, which can be thought of as resulting from some form of averaging over the microscopic structure. Bulk response functions can then, in turn, be deduced by utilizing a Maxwell-Garnett type averaging over these mesoscopic structures and employing the classical Mie theory of scattering of electromagnetic radiation.

The rather simple dependence of the bulk and mesoscopic properties on the response functions, that encapsulates the detailed quantum mechanical interactions, allows the design of model materials whose properties are determined by their nano structure i.e. metamaterials.

The proposed designs naturally need to satisfy certain constraints e.g. the parameters determining the bulk response functions should produce functions that are physically feasible and describe materials that can be fabricated by present or small extensions to existing techniques. This approach has been used to design Negative and Zero refractive Index Materials (NIM's [?] and ZIM's [?]), that have an extensive range of applications for instance from super lenses [?], small antenna [?] to medical imaging [?]. We have demonstrated that it is possible design Positive refractive Index Materials (PIM's) that can be tuned to be NIM's and ZIM's [?].

The observation that one can construct materials with desired responses to electromagnetic radiation has led to the exciting prospect of being able to design and manufacture cloaking devices [?]. These devices are based on the idea that they can deflect light around an object without loss or distortion and be effective over a broad band of frequencies. A major aim of this effort is to produce a device that will cloak an object from visible frequencies. In order to deflect radiation around an object the cloaking material needs to exhibit strong local variations in its response to light, while a precondition for a material to deflect without loss is that it should have maximal transparency [?] and minimal reflectivity.

In this letter we demonstrate that it is possible to design a material slab with bulk properties that produce a broadband transparency to optical frequencies, with very low reflectivity and absorption. Moreover we further demonstrate that this property can be switched off and on by, for example an external potential difference, which causes the transparent material to become a highly reflective opaque ZIM.

2. BACKGROUND

We consider a 500nm thick slab formed from randomly distributed identical nanospheres in a Liquid Crystal (LC) substrate that are composed of a hypothetical Drude material, which could be formed from a doped semiconductor [?]. We constrain the sizes of these nanospheres to be in the range of 15nm–100nm so that the slab could contain several spheres throughout its thickness and that the spheres are large enough that they essentially act as macroscopic particles with bulk like properties. By varying the doping we consider relative permittivities of the nanospheres to be in the range of 0.5 to -10.0. One could also consider solid metal spheres but then the choice of permittivities would be more limited. The permittivity of the LC substrate can be varied by applying a voltage across the ends of the slab [?].

The permittivity of the Drude nanosphere is described by the following expression

$$\varepsilon_1 = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega}, \quad (1)$$

where ω_p is the plasma frequency and γ is a damping term. The host material is assumed to be a LC whose relative permittivity ε_2 can be changed between 2 and 4. For linearly polarized electromagnetic radiation, incident as an extraordinary wave onto a nematic LC host, the relative permittivity is given by

$$\varepsilon_2 = \frac{\varepsilon_{||}\varepsilon_{\perp}}{\varepsilon_{||}\cos^2\theta + \varepsilon_{\perp}\sin^2\theta}, \quad (2)$$

where $\varepsilon_{||}$ and $\varepsilon_{\perp}$ are the permittivities for an electromagnetic wave polarized parallel and perpendicular to the director axis $\hat{n}$ respectively and θ is the angle between $\hat{n}$ and the incident light wave vector [?]. The director axis of the LC can be switched from parallel to perpendicular to the cell window either electrically (by an AC bias electric or magnetic field) or optically (through the optical intensity dependent director axis reorientation effect) [?]. We used generic values of the relative permittivity in the range of 2 to 4 to mimic the effect of this biasing. Finally, both materials (Drude nanosphere and LC host) are non-magnetic with relative permeability equal to 1.

The electromagnetic material properties of the slab composed of the LC substrate with the nanospheres randomly dispersed may be determined by employing the Maxwell-Garnet mixing rule [?][?]. The response of the bulk material is described by an effective permittivity

$$\varepsilon_{eff} = \varepsilon_2 \left(\frac{k^3 + 4\pi N a_1 i}{k^3 - 2\pi N a_1 i} \right), \quad (3)$$

and permeability

$$\mu_{eff} = \frac{k^3 + 4\pi N b_1 i}{k^3 - 2\pi N b_1 i}, \quad (4)$$

where N is the volume density of the spheres ($N = 3f/4\pi r^3$), f is the filling fraction of the composite and k is the wavenumber of the light in the LC medium. The Mie scattering coefficients a_1 and b_1 of the nano sphere correspond to the first-order multipole terms i.e., dipole terms [?].

We used the Genetic Algorithm (GA) to optimize the transmitted power for a LC substrate of relative permittivity 4, while at the same time minimizing the transmitted power for a LC substrate of relative permittivity 2, in order to obtain a significant tuning effect. The GA is a stochastic optimization technique [?][?] which is based on the Darwin's 'survival of the fittest' principle of evolution as its driving force and is a widely used and well accepted method for multi-parameter, multi-objective optimization problems. Here, a binary GA is utilized to optimize four parameters to obtain transparency at a pre-defined incident light frequency. These parameters are the radius of the solid nanosphere r ; the filling factor f ; the plasma frequency ω_p ; and the damping coefficient c_γ that determines the damping factor through

$$\gamma = \frac{\omega_p}{c_\gamma}. \quad (5)$$

In a binary GA, the optimization range is discretized by predefined bits. In this case, the GA optimized the radius in the range of 15nm to 100nm with 10 bits, the filling factor in the range of 0.2 to 0.5 with 3 bits, the plasma frequency in the range of $5.0265 \times 10^{14} Hz$ to $5.0265 \times 10^{16} Hz$ with 10 bits and the damping coefficient in the range 200 to 2000 with 10 bits. During the optimization the GA population size was 10, the maximum number of iterations was 150, the crossover probability was 0.5, and the mutation probability was 0.03. We used a dual fitness function

$$F(f, r, \omega_p, c_\gamma) = f_1 \delta_{\varepsilon_2 4} - f_2 \delta_{\varepsilon_2 2}, \quad (6)$$

composed of the optimal values of the fitness functions f_1 and f_2 , where

$$\begin{aligned} f_1 &= 5 \times [(Re(\varepsilon_{eff}) - 1)^2 + Im(\varepsilon_{eff})^2 + Re(\mu_{eff} - 1)^2 \\ &\quad + Im(\mu_{eff})^2] + |transmitted\ power(dB)|, \\ f_2 &= 30\Theta(|transmitted\ power| - 30dB) \\ &\quad + |transmitted\ power|\Theta(30dB - |transmitted\ power|), \end{aligned} \quad (7)$$

and $\Theta(x - y)$ is the Heavyside step function. This fitness function emphasizes strong transmission at $\varepsilon_2 = 4$ and low transmission at $\varepsilon_2 = 2$, reversing the strong and weak emphasize resulted in a fitness function that produced materials with high loss.

The standard Fresnel formulae (with loss) for light normally incident on a dielectric slab, see for instance [?], were used to compute the transmission, reflection and absorption of the slab formed by the liquid crystal and nanospheres.

3. RESULTS

We targeted four different incident frequencies, 414THz, 449THz, 550THz and 702THz, at which to optimize the transparency of the metamaterial slab for a LC with relative permittivity of 4, while at the same time producing a sharp tuning effect to non transparency, when the LC relative permittivity is changed to 2. The GA, with the fitness function shown in eq. 6, did not produce significant tuning over the visible range for the targeted incident frequency of 702THz, while the results for 414THz and 449THz showed a broader band transparency than the one for 550THz and are displayed in Fig.'s 1(a)–(c), 2(a)–(c), 3(a)–(b) and 4(a)–(b). All the optimizations yielded non magnetic materials, i.e. $\mu_{eff} = 1$, for incident light in the range 300–1000THz.

The optimization based on an incident light frequency of 414THz resulted in a filling factor $f = 0.350$, a plasma frequency of $\frac{\omega_p}{2\pi} = 706.5THz$, a sphere radius $r = 15nm$ and a damping coefficient $c_\gamma = 1998.1$. The results shown in Fig. 1(a)–(c) display that between 710nm and 820nm the transmitted power for the LC of relative permittivity of 4 is 85-90%, while the transmitted power for the LC of relative permittivity of 2 is 0-5%, which except for loss due to absorption at the edges of this range, corresponds to low reflectivity of the $\epsilon_2 = 4$ slab and high reflectivity of the $\epsilon_2 = 2$ slab. These results demonstrate that there is a switching from transparency to opaqueness in the red to very near infrared part of the spectrum as the relative permittivity of the LC changes from 4 to 2.

FIG. 1 (Color online). Plots of the magnitudes of (a) the power transmitted, (b) absorbed and (c) reflected by the light incident on the slab formed by a LC's that have relative permittivity of 2,3 and 4 for the optimization targeted for incident light of 414THz.

Both the slabs are transparent to visible wavelengths below 740nm. The $\epsilon_2 = 2$ slab shows high transparency, with a transmission power between 90 and 95%, while the $\epsilon_2 = 4$ slabs fluctuate but in general is still high though decreasing from 90% for red light to 70% for violet light. We have also shown the results based on a LC with relative permittivity of 3 to indicate that the dependence of the transparency to the variation of the relative permittivity is continuous.

The optimization based on an incident light frequency of 449THz produced a filling factor $f = 0.425$, a plasma frequency of $\frac{\omega_p}{2\pi} = 729.7THz$, a sphere radius $r = 15nm$ and a damping coefficient $c_\gamma = 1998.1$. The results shown in Fig. 2(a)–(c) exhibit similar behavior as the results of Fig. 1, but shifted to higher frequencies. They display that between 600nm and 700nm the transmitted power for the LC of relative permittivity of 4 is 80-85%, while the transmitted power for the LC of relative permittivity of 2 is 0-5%. Again, with the exception of loss due to

absorption at the edges of this range, this corresponds to low reflectivity of the $\epsilon_2 = 4$ slab and high reflectivity of the $\epsilon_2 = 2$ slab. The switching effect is now fully in the visible range between the red and orange bands.

FIG. 2 (Color online). Plots of the magnitudes of (a) the power transmitted, (b) absorbed and (c) reflected by the light incident on the slab formed by a LC's that have relative permittivity of 2,3 and 4 for the optimization targeted for incident light of 449THz.

Both of the slabs are transparent to visible wavelengths below 600nm. The $\epsilon_2 = 2$ slab shows high transparency, with a transmission power around 90%, while the $\epsilon_2 = 4$ slabs fluctuates but generally decreases from 83% for red light to 60% for violet light.

The variation of the real part of the refractive indices, shown in Fig.'s 3(a) and 4(a), over the visible spectrum is between 0 and 1.6, while the imaginary parts are nearly zero in the region where the light is transmitted, indicating that the material has very low loss. A more intriguing aspect of the results displayed in these figures is that at wavelengths longer than 690nm for the case optimized at 414THz and longer than 590nm for the 449THz case the real part of the refractive indices become very close to zero. The onset of this behavior occurs at different wavelengths for the LC's of different relative permittivity showing that this ZIM property is tunable.

FIG. 3 (Color Online) Plots of (a) the real and imaginary parts of the bulk refractive index and (b) of the real and imaginary parts of the surface impedance of the slab formed by a LC's that have relative permittivity of 2,3 and 4 for the optimization targeted for incident light of 414THz.

FIG 4 (Color Online) Plots of (a) the real and imaginary parts of the bulk refractive index and (b) of the real and imaginary parts of the surface impedance of the slab formed by a LC's that have relative permittivity of 2,3 and 4 for the optimization targeted for incident light of 449THz.

The ZIM behavior can be further analyzed by considering the reflectance and impedance plots Fig.'s 1(c), 2(c), 3(b) and 4(b) respectively, where one can observe that very small refractive indices correspond to reflected powers becoming very high and the real part of the impedances becoming zero.

4. DISCUSSION

In order to construct a cloaking device that functions in the optical range it must be transparent and bend the light around the object to be shielded without distortion or loss. We focused on the first condition and have shown that it is possible to design metamaterials that could feasibly be fabricated and are essentially transparent to a broad band of visible frequencies with low loss.

The structure that we consider is formed by randomly distributing solid nanospheres composed of a hypothetical Drude material that could be formed by a doped semiconductor [?] in a liquid crystal film forming a metamaterial slab. We assume that these spheres are large enough that their interaction with light can be described within the classical Drude picture using Mie theory and that the bulk optical properties of the metamaterial slab are determined by a Maxwell-Garnett averaging and limiting the description of light-sphere interaction to dipole terms. The nature of nanosphere material is established by a GA optimization procedure that determines its plasma frequency and damping constant for a targeted incident light frequency, while at the same time optimizing the size of the sphere and their density in the LC. Moreover by using a dual fitness function in the GA we have exhibited that this transparency can be switched on or off and even tuned by varying the permittivity of the LC substrate by, for example, applying a biasing voltage across the LC cell

The materials that we obtained through the GA optimization also became tunable ZIMs at red to near infrared frequencies, an important property that can be utilized in many areas e.g. more effective antennas and improved lenses. This ZIM behavior was seen to correspond to high reflectivity and very low real impedance.

The transmission power we obtained was between 80 and 95% over different parts of the visible spectrum and for LCs with different relative permittivity. Although this is not 100%, which would be desired for cloaking, it indicates that with improvements in the optimization technique and some modification of our design that this level could be approached. Another way to maximize the effectiveness of a given slab could be to tune the slab to optimally match the incident frequency in order to maximize transmission.

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